**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Truth Tables

**Name** _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

## Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. How can you recognise a NOT gate symbol?

2. 22 items fill boxes of 5. How many full boxes are there?

A) 4 B) 2
C) 5 D) 4.4

3. A bitmap is 60 x 20 pixels at 4 bits per pixel. Calculate the raw size in bytes, excluding metadata and compression.

4. Entry needs a valid ticket AND membership card. "AND is true if just one input is true." Correct this. What happens with only a ticket?

## Today's objectives

Write all four two-input combinations. Complete AND and OR columns without missing a row.

## Key vocabulary - Read, cover, recall

**Truth table**
Lists an output for every possible input case.

**Input row**
One combination of the Boolean inputs.

**AND**
Output 1 only when both inputs equal 1.

**OR**
Output 1 when at least one input equals 1.

**Write one key term and its meaning in your own words:**

| Do Now | I Do | We Do | You Do | Review & Improve |
|---|---|---|---|---|
| **Recall previous learning** | Read new knowledge | Practise with guidance | Work independently | Check my understanding |

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Learn and practise

## New learning - Key ideas

| | |
|---|---|
| **Four combinations for two inputs** | For A and B, list input pairs 00, 01, 10, 11 exactly once. Each row corresponds to one possible combination. |
| **AND truth table** | AND returns 0,0,0,1 for rows 00,01,10,11; only the final row has both inputs high. |
| **OR truth table** | OR returns 0,1,1,1; the first row is the only one without a high input. |
| **NOT truth table** | NOT has one input, so only two rows are required: 0 -> 1 and 1 -> 0. |

## I Do - One worked example

**Complete the AND output column for inputs 00,01,10,11, explaining the difference in row 11. [4 marks]**

1. Write rows 00,01,10,11 so no input case is missing.

2. 00 produces 0 [1] and 01 produces 0 [1].

3. 10 produces 0 [1], while 11 produces 1 [1].

4. Only 11 contains both inputs high, so it satisfies AND.

## We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Complete OR outputs for input pairs 00,01,10,11 in order. [4 marks]

2. Give NOT outputs for A=0 and A=1. [2 marks]

3. For A=1,B=0, explain AND and OR outputs. [4 marks]

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Think and write

## Code First - Predict > Write > Test > Improve

**YOUR TASK: Complete the condition that prints 1 only when A and B are both 1.**

```
for A = 0 to 1
for B = 0 to 1
if ... then
print(1)
else
print(0)
endif
next B
next A
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

## You Do - Independent exam questions

Q2. State the number of rows in a two-input truth table, excluding headings. [1 mark]

Q3. Complete the output column for an OR gate on input pairs 00,01,10,11. [4 marks]

Q4. Complete four-row AND and OR output columns and explain why the outputs differ for input 0,1. [6 marks]

## Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A two-input truth table has four combinations. ☐ T ☐ F
2. For A and B, list input pairs 00, 01, 10, 11 exactly once ☐ T ☐ F
3. AND returns 0,0,0,1 for rows 00,01,10,11; only the final row has both inputs high. ☐ T ☐ F

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

## Do Now - Answers

**Q1** It is a one-input inverter, usually a triangle with a small output bubble. ☐

**Q2** A - 4 ☐

**Q3** 60 x 20 x 4 = 4,800 bits; divide by 8 = 600 bytes. ☐

**Q4** AND is true only when both inputs are true. One valid ticket or card alone is not enough. OR needs at least one true input; NOT reverses a value. ☐

## Code First - Model

```
for A = 0 to 1
for B = 0 to 1
if A == 1 AND B == 1 then
print(1)
else
print(0)
endif
next B
next A
```

**Output: 0**
**0**
**0**
**1**

## We Do - Check the model

1. 00 -> 0 [1]; 01 -> 1 [1]; 10 -> 1 [1]; 11 -> 1 [1]. ☐

2. NOT 0=1 [1] and NOT 1=0 [1]. ☐

3. AND gives 0 [1] because both are not 1 [1]. OR gives 1 [1] because one input is 1 [1]. ☐

## Exit Ticket - Check

**1. TRUE 2. TRUE 3. TRUE**

## You Do - Self-mark

**Q2** Four rows [1]. ☐

**Q3** 0 [1]; 1 [1]; 1 [1]; 1 [1]. ☐

**Q4** AND 0,0,0,1 [2]; OR 0,1,1,1 [2]. For 0,1, AND is 0 [1] because both are not high, while OR is 1 [1] because one input is high. ☐

## Try the question again

**1. Copy the FULL question you found most difficult:**

**2. In GREEN PEN, write a complete new answer:**

| Do Now | I Do | We Do | You Do | Review & Improve |
|---|---|---|---|---|
| Recall previous learning | Read new knowledge | Practise with guidance | Work independently | **Check my understanding** |